

B. Change it up!



Time Constraints

- **java** : 3.0 Seconds



Memory Constraints

- **java** : 256MB

In order to enhance the safety of traveling and alleviate traffic congestion, one-way traffic is now implemented worldwide. The interim government of Bangladesh decided to remain updated with emerging trends. In the past, the ring-shaped network of R two-way roads connected all R cities of Bangladesh. This meant that each city was directly connected to exactly two other cities, and it was possible to travel to any other city from any of the R cities. After the introduction of one-way traffic on all R roads by the Government of Bangladesh, it was soon obvious that it was impossible to travel from certain cities to others. Now for each road, it is known in which direction the traffic is directed at it, and the cost of redirecting the traffic. What is the minimum amount of money that the government should allocate to the redirection of roads to ensure that it is possible to travel from every city to another?

Input

- First line: integer R (number of cities/roads), ($3 \leq R \leq 100$)
- Next R lines: three integers (a_i, b_i, c_i) (road from (a_i) to (b_i) with redirection cost (c_i))

Output

Output a single integer: minimum cost to make the graph strongly connected.

Sample Input 1

```
3
1 3 1
1 2 1
3 2 1
```

Sample Output 1

```
1
```

Sample Input 2

```
3
1 3 1
1 2 5
3 2 1
```

Sample Output 2

```
2
```

Sample Input 3

```
6
1 5 4
5 3 8
2 4 15
1 6 16
2 3 23
4 6 42
```

Sample Output 3